



Learning Python - Sheet #11

Spacewar #2

Now it's time for our spaceship to shoot!



First, let's download the picture of a projectile. Name it **spaceship.png**. For example, you can use this one: https://www.flaticon.com/free-icon/fireball_7734417

Then update your file (**spacewar.py**) to add the following lines in the top part, before the functions definition.

```
missile = pygame.image.load('fireball.png')
# Make the picture smaller
missile = pygame.transform.scale(missile, (16, 16))
reload_in = 0

RELOAD_DELAY = 20
V_MISSILE = 10
V_SPACESHIP = 5
```

Now add the following functions to your code. These are functions to handle creating and moving the missiles.

```
def fire_missiles():
    global reload_in
    if reload_in == 0:
        pressed = pygame.key.get_pressed()
        if pressed[pygame.K_SPACE]:
            reload_in = RELOAD_DELAY
            # Positions of canons in (512, 512) spaceship:
            # (360, 25), (480, 105), (480, 370), (360, 450)
            # We're placing a (16, 16) missile sprite, so shifting y by ~-8
            # Scaling the result for a (64, 64) spaceship.
            missiles.append((x+43, y-2))
            missiles.append((x+43, y+50))
            missiles.append((x+60, y+8))
            missiles.append((x+60, y+42))
    else:
        reload_in -= 1

def move_missiles():
    for (i, m) in enumerate(missiles):
        missiles[i] = (m[0] + V_MISSILE, m[1])
```



Pay attention: Note how we create 4 missiles. That's because our spaceship has 4 cannons. We use some magic numbers to specify where the missiles are created relative to the spaceship. These are magic numbers. Normally we'd prefer moving them to some variable and give them a clear name (for example: CANON1_X, CANON1_Y, etc ...)

We also force a delay between two salvos of missiles by setting `reload_in` to `RELOAD_DELAY` and decrementing `reload_in` once per frame if it's not 0. You can only fire if `reload_in` is 0.

Finally, change the function `draw` and your game loop (add the calls to the new functions and draw the missiles):

```
def draw():
    screen.fill("black")
    screen.blit(spaceship, (x, y))
    for m in missiles:
        screen.blit(missile, m)
    pygame.display.flip()

while running:
    handle_events()
    move_spaceship()
    fire_missiles()
    move_missiles()
    draw()
    clock.tick(60)
```

